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The nature and structure of European belief systems: exploring the varieties of belief systems across 23 European countries

Jochem van Noord[1,2,*], Felicity M. Turner-Zwinkels[3], Rebekka Kesberg[4,5], Mark J. Brandt[6], Matthew J. Easterbrook[4], Toon Kuppens[7] and Bram Spruyt[2]

[1]Department of Social Psychology, University of Groningen, 9712 TS Groningen, The Netherlands
[2]Brussels Institute of Social and Population Studies (BRISPO; Research cluster TOR), Vrije Universiteit Brussel, 1050 Elsene, Belgium
[3]Department of Sociology, Tilburg University, 5000 LE Tilburg, The Netherlands
[4]School of Psychology, University of Sussex, Falmer, Brighton, East Sussex, BN1 9QH, United Kingdom
[5]Political Science Department, University of Amsterdam, Nieuwe Achtergracht 166, 1018 WV Amsterdam, The Netherlands
[6]Department of Psychology, Michigan State University, 48824 East Lansing, USA
[7]Department of Organizational Psychology, University of Groningen, 9712 TS Groningen, The Netherlands

[*]Corresponding author. Email: j.van.noord@rug.nl

We investigate the structure of political belief systems across Europe to investigate what belief systems in European societies, and those who hold them, have in common. In doing so, we answer three questions: First, are political belief system structures similar across Europe? Second, which demographic groups are likely to have similar belief systems within countries? Third, how are belief systems related to voting behaviour? Results from Correlational Class Analyses on data from 23 European countries indicate that a wide variety of belief systems exist in Europe (2–5 per country), but that these can be summarized into two diverse groups, although belief systems in one group were more similar than in the other. Unexpectedly, the groups did not differ in the strength of association between beliefs. While cultural and economic belief dimensions were not consistently found, and tended to be weak, they were positively associated in the first group and negatively associated in the second. Belief systems of the first group were more likely to be from Western European countries and its members more likely to be higher educated compared to the second group. Membership in the second group was associated with more populist far-right voting and vote abstention.

Political beliefs do not exist in isolation. They are interrelated elements of (implicit) belief systems, ideologies, or discourses (Converse, 1964; Gerring, 1997). Although that insight has long been present in the literature on public opinion, recent methodological advances have made it easier to investigate belief systems as a relational *network* (Borsboom *et al.*, 2021). In such a network, nodes represent beliefs and the edges or links between nodes are the associations between beliefs (Boutyline and Vaisey, 2017; Brandt, Sibley and Osborne, 2019). This network approach has made at least two important contributions. First, it has made it possible to identify belief systems from the bottom-up, rather than testing, top-down, how well pre-defined belief systems fit a specific group of people. This allows researchers to forego any preconceived belief systems and explores how citizens themselves structure their beliefs. Second, studying beliefs as part of belief *systems* allows insight into the way beliefs relate to one another. In this way, it shifts the focus from *what* people think to *how* people think (Baldassarri and Goldberg, 2014; Daenekindt, De Koster and Van der Waal, 2017; DellaPosta, 2020). Taking into account the fact that different sections of the public may have different belief systems (and some might not have any), analysing belief systems helps us to understand the structure of current political debates.

Notwithstanding the progress that has been made in empirical work on belief systems, research has until now predominantly focussed on single countries, most

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often the United States (e.g., Baldassarri and Goldberg, 2014; Boutyline and Vaisey, 2017; Brandt, Sibley and Osborne, 2019; but see Barbet, 2020; Gonthier and Guerra, 2022; Keskintürk, 2022). As a consequence, we have limited insight into whether and how belief systems vary between countries, rather than solely within countries. Europe is a particularly interesting context in which to test for similarity in between-country political belief systems. This is because, on the one hand, many different political landscapes exist across Europe—each country has its own political system with a different number of parties, the level of party competition and polarization, and the ideological variation and success of these parties. On the other hand, these countries are strongly connected through shared history and institutions (e.g., the EU and other European institutions), and increasingly so, with political parties and movements operating on a European level, rather than on solely the national. Against that background, we seek to address the overarching research question of what do belief systems in European societies, and those who hold them, have in common? We do this by asking three sub-questions. First, are the structures of political belief systems similar *across* countries in Europe? Second, which demographic groups are likely to have similar belief systems *within* countries? Third, how are different belief systems related to political behaviour (voting)?

By answering these questions, we make two contributions. First, we describe how similarly structured and interrelated each belief system is within the space of European belief systems without presupposing a single European belief system nor a single belief system for each country. We do this by exploring the belief systems of each country separately and then comparing these resulting belief systems across countries. We use data from the European Social Survey to investigate variation in the structure of political belief systems within and between 23 European countries ($N = 37{,}118$). By testing the broader system of belief system networks in Europe we offer fresh insight into the extent that political belief systems are similar or different across Europe and thereby, in the context of increasing European political integration, into how public debate across Europe is structured. Cross-national similarity in how people think about politics revolve fundamentally around the existence or possibility of common ways of thinking and talking about politics. Differences in belief systems would affect the extent (political) actors (politicians, political movements, activists, etc.) might struggle to communicate broader political discourses and programs effectively and struggle to build coalitions across countries.

Second, we explore the behavioural consequences of belief systems by examining how the belief systems we find are related to voting behaviour. Belief system structure has been associated with voting behaviour in the Netherlands (Daenekindt, De Koster and Van der Waal, 2017), suggesting that the structure of belief systems is important for political decision-making. If this relation between-belief systems and voting behaviour extends broadly across Europe, then this would demonstrate that these belief systems have robust behavioural relevance, but also connect them to current European political trends.

Possible patterns in the structure of belief systems in Europe

A belief system is the 'configuration of ideas and attitudes in which the elements are bound together by some form of constraint or functional interdependence' (Converse, 1964: p. 3; Gerring, 1997; cf. Brandt and Sleegers, 2021). Belief systems thus describe *how* beliefs are related to each other, *but not* the extent to which the beliefs are endorsed. Thus, people with strongly different political beliefs might have similar belief systems, due to similarities in how the beliefs relate to each other (Boutyline, 2017). We assume that within any one country, multiple political belief systems are likely to exist (Achterberg and Houtman, 2009; Baldassarri and Goldberg, 2014; Barbet, 2020; Gidron, 2020). Different cultural backgrounds, socialization, media diets, and group interests are likely to influence political interests and connect associated attitudes in different ways for different groups of people, resulting in differently structured political belief systems (Baldassarri and Goldberg, 2014; DiMaggio *et al.*, 2018; Goldberg and Stein, 2018; Groenendyk, Kimbrough and Pickup, 2020; Brandt and Sleegers, 2021; Brandt, 2022). Previous research has shown that belief systems can be distinguished from each other in at least two key ways: (i) the extent of *constraint* (structured or unstructured) and (ii) its dimensionality (one- or multi-dimensional). Variation in these two key characteristics of belief systems should ultimately produce different belief system structures. We will review this first below, then we will review how we expect these to differ across Europe.

Structured and unstructured belief systems

According to Converse (1964; see also Gerring, 1997), belief systems have 'constraint' such that it should be possible to predict an individual's stance on one political attitude from their stance on another attitude. The idea was that if the key structuring principle was political ideology, then it should be possible to make such predictions. That is, someone who holds, for example, conservative attitudes on gender should be more likely to hold conservative attitudes on welfare. However, Converse showed that most members of the mass public had only weak correlations among their political beliefs. In other words, they did not have a structured belief system (see also Kinder, 2006; Baldassarri and Gelman, 2008). Highly structured belief systems, with strong correlations between political beliefs, were only found among a political elite that he named 'ideologues' (Converse, 1964: p. 13).

There is, indeed, empirical evidence that at least some members of the public do have (tightly) structured belief systems (Stimson, 1975; Groenendyk, Kimbrough and Pickup, 2020). Such high levels of constraint and the associated tightly structured belief systems are especially likely to be found among people with more political interest and engagement, including political elites and people with higher levels of political knowledge (Jennings, 1992; Lupton, Myers and Thornton, 2015; Fishman and Davis, 2022). In contrast, low levels of constraint (i.e., an unstructured belief system) are found in people with low levels of political knowledge or interest. It appears that their lower familiarity with politics leads people to organize their belief systems in more idiosyncratic ways and with greater measurement error (Converse, 1980; Judd and Milburn, 1980). This unstructured pattern has also been found in Europe (Daenekindt, De Koster and Van der Waal, 2017; Gidron, 2020). European research suggests that belief systems in Europe are marked by various levels of constraint, and can therefore be seen as more or less structured or even unstructured. On a continuum, belief systems would range from (i) a structured, *ideologue* belief system with *high correlations* among political beliefs (i.e., beliefs are consistently left/right-wing), mostly shared by individuals with higher levels of political knowledge and/or political elites, and (ii) an *unstructured* belief system with *weak or no discernable correlations* among political beliefs mostly shared by those with lower levels of political knowledge and interest.

Multi-dimensional belief systems

A belief system that is made up of perfectly consistent political beliefs (e.g., political attitudes are either all left-wing or all right-wing) will be one-dimensional. However, political belief systems often have more than one dimension (Gidron, 2020; Treier and Hillygus, 2009; Saucier, 2000; for a review see Johnston and Ollerenshaw, 2020). Research has consistently found that political beliefs are structured both along an economic and a cultural dimension, where the former tends to unite beliefs around issues of economic freedom and redistribution, the latter around morality and social freedom (Lipset, 1959; Middendorp, 1991; Van der Brug and Van Spanje, 2009; Johnston, Lavine and Federico, 2017). Among higher status members of the population (e.g., highly educated, politicians), cultural and economic attitudes 'go together' in such a way that they align along a single dimension (Achterberg and Houtman, 2006; Baldassarri and Goldberg, 2014; Alexandre, Gonthier and Guerra, 2021). This would be the case for the structured, *ideologue* belief system described above (cf. Converse, 1964; Baldassarri and Goldberg, 2014). However, for others, these two dimensions can be correlated negatively, so that people who share this belief system would be left-wing economically and right-wing culturally (or vice versa), resulting in a two-dimensional belief system (Baldassarri and Goldberg, 2014; Johnston, Lavine and Federico, 2017). Evidence for both the *ideologue* and this *two-dimensional cultural-economic* belief systems has been found widely across Europe (Achterberg and Houtman, 2009; Gidron, 2020). Importantly, the two-dimensional cultural-economic belief system should not be thought of as unstructured. This is because deviations from the ideologue belief system that result in the alternative (two-dimensional cultural-economic) belief system are not necessarily caused by a lack of political knowledge, and may instead be motivated by specific group interests (Achterberg and Houtman, 2009; Elchardus and Spruyt, 2012; Malka, Lelkes and Soto, 2019; Groenendyk, Kimbrough and Pickup, 2020).

There are also two-dimensional belief systems not structured around economic versus cultural dimensions (e.g., Pless, Tromp and Houtman, 2020). However, research on two-dimensional belief systems tends to focus on a small number of attitudes that are selected because they fit along the dimensions under study. Measuring a broader range of beliefs, as we do, might uncover belief systems with more than two dimensions.

The previous arguments on both structuredness and dimensionality suggest that we may find at least three general types of belief systems: (i) *Ideologues,* who have a one-dimensional belief system (that is consistently left or right wing) that is structured by (predominantly) strong positive correlations among beliefs. (ii) A group of belief systems that are *alternative* to the ideologue system by deviating from the one-dimensional left-right structure. The most typical example of such an alternative belief system in the literature is the *two-dimensional cultural-economic system*, where the subgroups of cultural or economic beliefs correlate positively, but these subgroups do not correlate positively among each other. Lastly, (iii) a group of *unstructured* belief systems that have weak and/or inconsistent correlations between beliefs (i.e., low constraint, so being left-wing on one belief does not imply being left-wing on another).

Similarities and differences in belief systems across countries

Although we expect these (ideologue, alternative, and unstructured) belief systems to be present across Europe, the extent that each belief system resembles its prototype is likely to vary. If so, we will find variation in how similar or prevalent these belief systems are across countries. Indeed, each country has its own particular political history, issues, and concerns that are uniquely salient (Carmines and Stimson, 1986, 1993; Green-Pedersen, 2007). Different countries also have different political systems (Keskintürk, 2022) and different cultural cleavages that may structure belief systems in unique ways. This will result in variation in the number, strength, and direction of associations between beliefs in a belief system (Malka, Lelkes and Soto, 2019; Keskintürk, 2022). Nevertheless, at the same time, European countries share a political system (European Union) and political parties (in the European Parliament) are members of formal party families and fractions (e.g., European People's Party and Identity and Democracy). Even European countries outside of the EU are culturally and economically interdependent with the rest of Europe. Although such interdependence has not led to indiscriminate convergence of political beliefs, at least some convergence in attitude contents has been found (Akaliyski, 2019). Such similarities and interdependence may likewise contribute to similarities in belief system structure.

The political belief systems of ideologues should be relatively common across countries in Europe because they are thought to stem from similar experiences across European countries. Specifically, European countries tend to be democratic and their political elites tend to have a left-right organization of political beliefs (Huber and Inglehart, 1995; Caprara and Vecchione, 2018). Thus, we expect the ideologue belief system to be present across countries. Similarly, the unstructured political belief system should also occur broadly across Europe, as each country is likely to have a group of people uninterested in politics with low levels of political knowledge. However, given the diversity of possible alternative belief systems, it is not straightforward to predict their emergence across Europe. To the extent a specific alternative belief system emerges due to specific societal events experienced across Europe, this alternative belief system will emerge in all such societal contexts. Research on the effects of globalization on dealignment/realignment of political landscapes indicates that such common societal events—such as economic globalization and migration—do exist (Kriesi et al., 2008). For example, it is possible that the *two-dimensional cultural-economic* belief system is more prevalent in Eastern than in Western Europe, due to the shared political history of communism in Eastern Europe (Malka, Lelkes and Soto, 2019). So, although we expect to find three or more broad groups of political belief system structures in Europe, the exact structure of these political belief systems and their prevalence should vary between countries.

Who is likely to have what belief system?

Historically, people's group memberships (e.g., social demographics and especially people's class position and religion) have been intertwined with their political belief systems. A reason for this is that the structure of belief systems is determined by normative consensus within groups (Groenendyk, Kimbrough and Pickup, 2020). In other words, there is no 'right' way for belief systems to be organized. Groups of people develop the structure of the belief system that they jointly agree on. This means that belief systems are at least in part determined by important group interests and needs. If a sufficiently large group of people share the same interests, this will be expressed in the emergence of a specific belief system structure associated with this group membership. As such, while belief systems exist 'in' an individual's mind (Brandt, 2022), it is also a social and collective construction, and hence, it is likely structured by social demographics.

Although prior research found that class/income (Baldassari and Goldberg, 2014; Gidron, 2020), and religion (Pless, Tromp and Houtman, 2020) are related to belief system structure, the (relative) organization of European politics around these two pillars has been in decline through the latter half of the twentieth century (Inglehart and Rabier, 1986; Pless, Tromp and Houtman, 2020). At the same time, political divides marked by education differentials have grown (Pless, Tromp and Houtman, 2020). In line with this, education level was recently found to be a strong predictor of how some Europeans (e.g., Dutch, Belgian) organize their political attitudes (Achterberg and Houtman, 2009; Elchardus and Spruyt, 2012). Higher educated people typically endorse a one-dimensional political belief system (similar to political elites; Van der Brug and Van Spanje, 2009), whereas less educated people typically endorse a two-dimensional cultural-economic political belief system.

There are at least two reasons for this educational difference in belief system structure. First, educational groups share a common experience. Political elites are usually higher educated (Bovens and Wille, 2017; Turner-Zwinkels and Mills, 2020). As such, higher educated members of society may be socialized into the same belief systems due to their shared cultural background (Noordzij, De Koster and Van der Waal, 2021). Second, educational groups share group interests. People with lower education levels are more likely to be vulnerable members of society (Elchardus and Spruyt, 2012) who experience relative deprivation, are seen as low-status members of society (Van Noord *et al.*, 2019, 2023), and may be exposed to negative discrimination and derogation (Gidron and Hall, 2017, 2019; Kuppens *et al.*, 2018). Their vulnerable status in society pushes them simultaneously to support right-wing cultural attitudes (which seek to protect them from cultural threats) and left-wing economic attitudes (which seek to protect them from economic hardship). As such, we expect that those with higher education levels will tend to have ideologue belief systems, and those with lower education levels to an alternative (two-dimensional cultural-economic) belief system.

Are belief systems related to voting behaviour?

To investigate the political relevance of belief systems for understanding politics, we explore to what extent belief system membership is related to voting behaviour. Previous research in The Netherlands has shown that belief systems containing only moral and cultural issues are related to voting behaviour (Daenekindt, De Koster and Van der Waal, 2017). The relation between voting behaviour and belief systems is likely to reflect both how parties and party elites shape belief systems and how belief systems motivate voters towards voting for specific parties (Groenendyk, Kimbrough and Pickup, 2020). In any case, it is likely that belief systems' political relevance depends partially on its relation to voting behaviour such that it is not merely a way of illustrating the connection between beliefs, but that also *how* people think has a relationship with relevant political cleavages.

We test this by analyzing the relationship between-belief system membership and voting behaviour towards three kinds of voting: loyalty, voice, and exit (cf. Hirschman, 1970). *Loyalty voting* means voting for a mainstream party, *exit voting* means vote abstention. *Voice voting* means voting for a non-mainstream party: far left, far right, populist, or a combination of these (i.e., populist far-right and populist far-left). This allows us to connect belief systems to political debates about the democratic resentment that lead people to vote against the political establishment, or refrain from voting entirely (Daenekindt, De Koster and Van der Waal, 2017; Kemmers, 2017; Gidron and Hall, 2019), but also to distinguish between different strands of 'voice voting'. The surge of populist radical right parties in the past 20 years, has (increasingly) been a Europe-wide phenomenon, with radical right-wing parties having electoral success in many European countries. Our analyses allow us to investigate to what extent differences in belief systems are related to voting for radical right-wing populist parties, but also whether these differences in belief systems extend beyond this particular form of (anti-establishment) voice voting.

Study overview

We aim to address three key questions: First, are the structures of political belief systems similar across countries in Europe? In order to address this question, we will explore how the belief systems we find in European countries are structured and how this differs across Europe. Second, we will investigate who shares what belief system? In other words, we will explore whether the belief systems are associated with certain demographic groups (e.g., education, income, religiosity). Finally, we investigate whether belief system membership is related to political behaviour. In the next section, we will explain our analytical strategy, before delving into the results.

Analytical strategy

We analyse the 2016 wave of the European Social Survey (NSD, 2016; see Table 1).[1] We chose this wave of ESS because of its broad range of relevant items.[2] From this survey, we selected 20 political beliefs to estimate political belief systems. These political beliefs are all beliefs about political systems, institutions, and policies that are embedded within a larger ideology or narrative about society or social groups. While most of these belief variables are single items, some are scales (scores are averages across all items available in the data). We use scales to ensure that the relative weight (compared to single items) of these concepts is not artificially increased by using multiple items for one concept. The nature of these beliefs is broad (including beliefs that are not easily categorized as 'cultural' or economic') and, in line with earlier research on belief systems (Ellis and Stimson, 2012; Brandt, Sibley and Osborne, 2019), consist of both *operational* components (i.e., specific policy stances) and *symbolic* components (e.g., political left-right identification). Full information on these belief variables can be found in the Supplementary Material (Supplementary Tables A1–A3). We coded all beliefs such that higher scores indicate right-wing responses, and lower scores indicate left-wing responses according to a traditional one-dimensional left-right belief system.

In order to identify belief systems within each European country, we used *correlational class analysis* (CCA, Boutyline 2017) for each country. This

Table 1 Description of all included beliefs

Beliefs	Description/example item	Scale info
Left–right identification	'In politics people sometimes talk of "left" and "right". Where would you place yourself on this scale?'	
Gender inequality	'When jobs are scarce, men should have more right to a job than women'	
Anti-LGBT	'Gay men and lesbians should be free to live their own life as they wish' (*r*)	$n = 3$; $\alpha = 0.80$
Euroscepticism	'Some say European unification should go further. Others say it has already gone too far'.	
Anti-immigration	'To what extent do you think [country] should allow people of the same race or ethnic group as most of [country]'s people to come and live here?'	$n = 3$; $\alpha = 0.87$
Anti-egalitarianism	'The government should take measures to reduce differences in income levels' (*r*)	$n = 3$; $\alpha = 0.59$
Benefits harm economy	Perception of negative economic effects of social benefits 'Social benefits and services in [country] place too great a strain on the economy'	$n = 2$; r = 0.48
Benefits harm society	Perception of negative social effects of social benefits 'Social benefits and services in [country] prevent widespread poverty' (*r*)	$n = 2$; $r = 0.53$
Welfare chauvinism	'Thinking of people coming to live in [country] from other countries, when do you think they should obtain the same rights to social benefits and services as citizens already living here?'	
Anti-economic interventionism	No government responsibility to provide standard of living 'It should not be governments' responsibility at all to (…) ensure a reasonable standard of living for the old'	$n = 3$; $\alpha = 0.68$
Anti-social benefits (low income)	'(…) government providing social benefits and services only for people with the lowest incomes, while people with middle and higher incomes are responsible for themselves?' (*r*)	$n = 3$; $\alpha = 0.68$
Anti-social benefits (parents)	'(…) government introducing extra social benefits and services to make it easier for working parents to combine work and family life even if it means much higher taxes for all?' (*r*)	
Educational spending	'Would you be against or in favour of the government spending more on education and training programs for the unemployed at the cost of reducing unemployment benefit?'	
Anti-basic income	'Some countries are currently talking about introducing a basic income scheme. Overall, would you be against or in favour of having this scheme in [country]?'	
Anti-climate change taxes	'Increasing taxes on fossil fuels, such as oil, gas and coal' (*r*)	
Anti-climate change renewables	'Using public money to subsidise renewable energy such as wind and solar power' (*r*)	
Anti-climate ban of appliances	'A law banning the sale of the least energy efficient household appliances' (*r*)	
Climate scepticism	'Do you think that climate change is caused by natural processes, human activity, or both?'	
Authoritarianism	'Important to live in secure and safe surroundings'	$n = 5$; $\alpha = 0.70$
Anti-libertarianism	'Important to try new and different things in life' (*r*)	$n = 5$; $\alpha = 0.66$

Note: all beliefs coded so that higher scores mean more right-wing/conservative answers. Description contains an example item in quotation marks (text taken from British questionnaire). (*r*) indicates that the item is reverse coded, so that disagreement means higher scores. Scale info indicates number of items and reliability (Pearson correlation with two items, Cronbach's alpha with two or more items). Supplementary Table A1 lists the ESS item names that were used to construct the belief (scales). Supplementary Tables A2 and A3 list descriptive statistics.

method clusters individuals based on how a range of variables *relate to each other,* rather than on the mean values people have for these variables. This is done by comparing the similarities between the *response patterns* of all respondents and then clustering the respondents based on these similarities. This identifies classes that include respondents with different values on these variables, but where all respondents are (relatively) similar in how the variables are correlated with each other—hence a *correlational class*. Formulated differently, a correlational class groups people who may hold different views on the world, but for whom the correlations between these views are similar. CCA differs from factor analysis, as, while both deal with interrelations between sets of variables, CCA does not find latent variables that explain common variance. CCA aims to partition a sample into subgroups with similar relations between beliefs. As such, it is also different from conventional latent class analysis, which also partitions a sample into subgroups, but does so on the basis of the stances (scores/values) on these beliefs, rather than the (cor)relations between beliefs.[3] This makes CCA an ideal method to answer our research question because it is a data-driven method that flexibly detects multi-dimensional belief system structures shared among groups of people (Boutyline, 2017; DiMaggio *et al.*, 2018; Barbet, 2020; Boutyline and Soter, 2020). This allows us to identify which belief systems emerge, rather than whether or not our researcher-imposed belief systems fit the population well.

We modified the existing CCA algorithm in two ways. First, CCA results can differ depending on the direction that the beliefs are coded. To circumvent this, we calculate correlations between individual response patterns 1,000 times (the first step of the CCA algorithm), each time reverse coding a random subset of a random number of beliefs, and then take the mean correlation of these iterations. This modified approach provides a more stable result than that of the original method. Second, the standard CCA algorithm does not allow for individuals with any missing values. We adjusted the CCA procedure to allow individuals to have a maximum of two missing values. Using this modified method, the total number of respondents is 37,118 (from 25,762, raw $N = 44{,}387$). For more (technical) information on and reasons for these adjustments, see Supplementary Appendix 2.

The analysis proceeded in several steps. We have also illustrated the different analyses and how the results of these analyses are connected to each other in Figure 1. The first step is the CCA, that, as explained above, finds clusters in a network of individuals (Figure 1A) based on the similarity in their response patterns. We perform this analysis on each country separately. This approach accommodates the multilevel structure of the data, but also departs from the notion that nation states (national institutions, political systems, etc.) are still the main political frame of reference for most individuals. Another advantage of this approach is that it allows us to uncover differences within certain broader types of belief systems across different countries.[4] For each correlational class a correlation matrix is produced that contains all correlations between all the included beliefs. This matrix of belief-correlations can be presented as a network of beliefs (Figure 1B), where beliefs are connected through their correlations (*within*-belief system correlations).

In the next step, we calculated the similarity between each belief system across all of the countries. This was done by calculating a Pearson correlation coefficient between all the (unique) within-belief system correlations of a belief system with each other belief system. These *between*-belief system correlations can be presented as a network of belief systems (Figure 1C). We used a clustering algorithm (leading eigenvector algorithm; Newman, 2006) to find clusters or groups of similar belief systems. We use these groups of belief systems to structure our analyses below on whether the structures of political belief systems are similar across countries in Europe and focus on both the differences between and within these groups of belief systems. These analyses focus primarily on density (constraint) and dimensionality of the belief systems. After these analyses on the structure of belief systems, we will explore how membership in these belief systems are related to demographics (i.e., who shares what belief system?) and how belief system membership is related to voting behaviour.

Results

Which European belief systems do we find and how are they structured?

The CCA analyses (Figure 1A) identify 70 belief systems[5] over the 23 countries (range [2,5] per country).[6] These 70 belief systems are in Figure 1B. In Supplementary Table A5 we list all belief systems and their descriptive statistics. To help us understand how similar the belief systems are to one another we computed Pearson correlations between the 70 belief systems (Figure 1C). These between-belief system correlations tell us how similar or how different the strength of correlations in two belief systems are and thus how similar or how different the belief systems are from one another.[7] The between-belief system correlations range from −0.053 to 0.792, with a mean correlation of 0.326. Hence, these correlations demonstrate that substantial differences exist between some of the belief systems, but also that other belief systems are relatively similar.

Next, we generated a network of belief systems (Figure 1C), which is presented in Figure 2. We then use a clustering algorithm to find clusters or groups of belief systems that are similar. This process helps us to understand which belief systems are more similar to

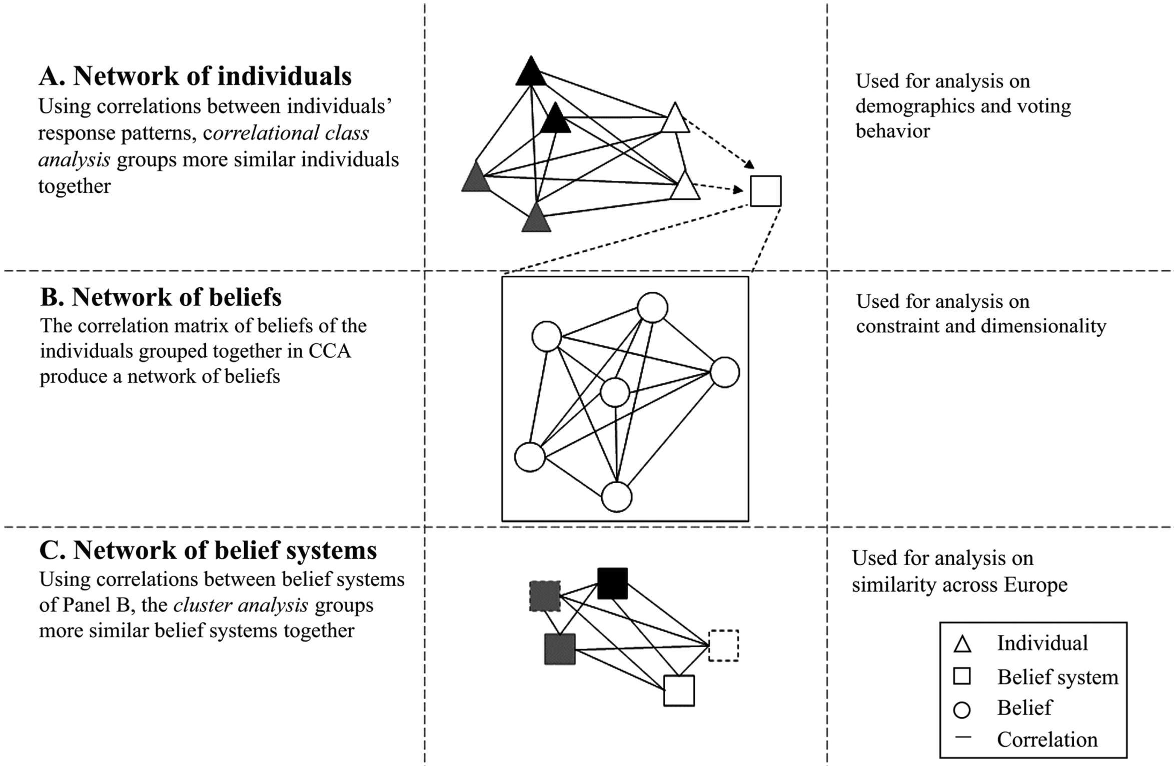

Figure 1 Analytical overview of correlational class analysis, cluster analysis, and belief systems

Note: All networks displayed here are example networks. Legend on the bottom left. Similar colours refer to similar (A) belief systems or (C) groups of belief systems. In the Network of belief systems, the belief systems with a dashed outline are example belief systems of a different country, that are grouped together with other belief systems through between-belief system correlations. In the case of the analysis on demographics and voting behaviour, we use data on membership of belief systems.

one another and guides our further analyses. The clustering algorithm found two belief system groups.[8] This suggests there are two broad types of belief systems in Europe. The dense edges among the belief systems in Group 1 (i.e., the white nodes) indicate that these belief systems are much more similar to each other than the (more spaced out) belief systems in Group 2 (i.e., the grey nodes). Indeed, the between-belief system correlations for Group 1 are about 1.8 times stronger than the between-belief system correlations for Group 2 (see Table 2). As such, it seems that Group 1 belief systems are more similar than Group 2.

Constraint is often a primary difference between-belief systems. We use within-belief system density (i.e., mean absolute strength of the correlation between beliefs) as a measure of constraint (Fishman and Davis, 2022). The belief systems of Group 1 have a similar mean within-belief system density as those in Group 2 (see Table 2). Thus, the mean absolute strength of correlation of these belief systems is roughly equal between the groups ($t_{Welch}(67.89) = 0.32$, $P = 0.75$). As Figure 2 shows, variation in density exists more *within* the groups rather than *between* the groups. The key finding here is that the main difference between Group 1 and Group 2 is not a difference in constraint, contrary to expectations. To further explore what the primary differences between these groups are, we look at the structure, specifically the dimensionality, of the belief systems. To do this, we explore three questions: (i) how many dimensions are there? (ii) What are these dimensions? And (iii) How do these dimensions relate to each other?

Table 2 Descriptive statistics on Group 1 and Group 2 belief system groups

Descriptive statistic	Group 1	Group 2
Number of belief systems	32	38
Mean between-belief system correlation	0.509	0.281
Mean of within-belief system density	0.092	0.090

Note: Group 1 is white, and Group 2 is grey in Figure 1.

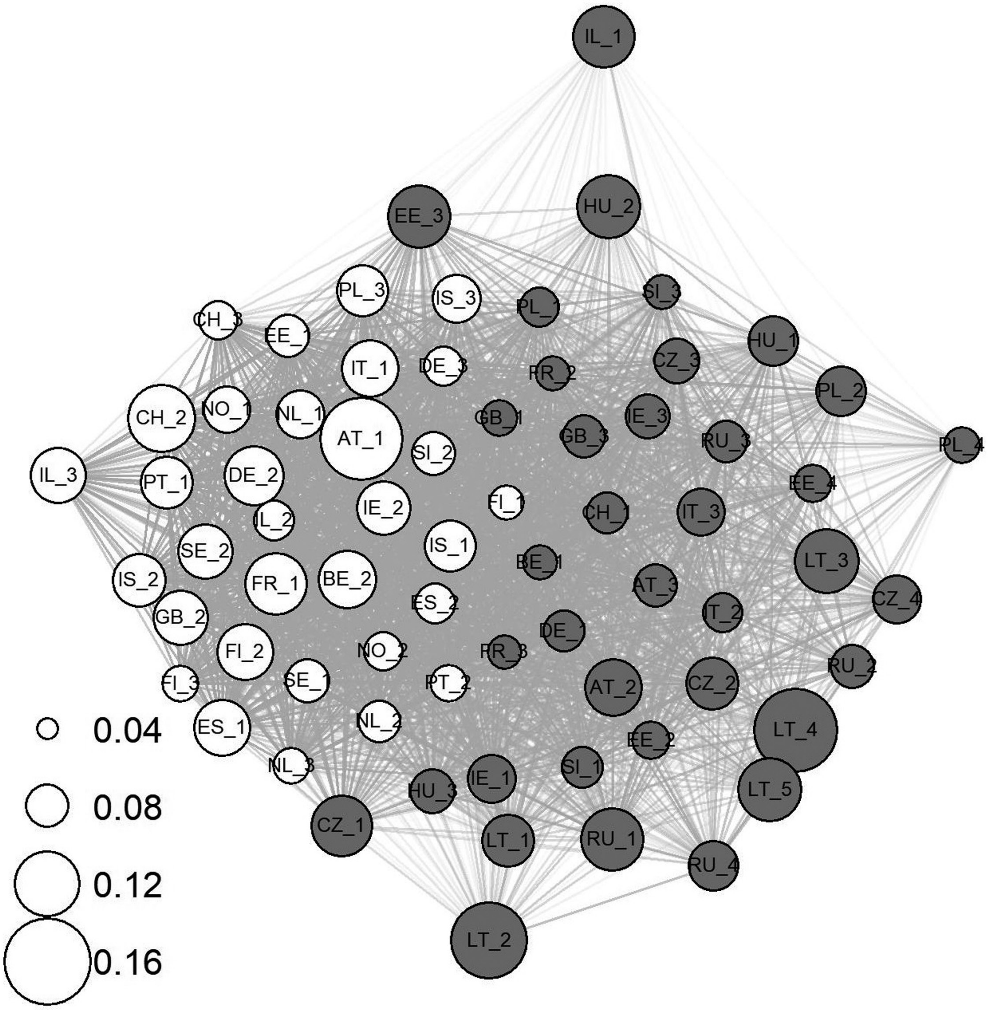

Figure 2 Network of all included belief systems (nodes scaled by density)

Note: Edges are Pearson correlations between-belief systems indicating their similarity. Only correlations higher than the mean correlation are shown, more opaque lines mean stronger correlations. Group 1 belief systems in white, Group 2 belief systems in grey. Nodes are scaled by within-belief system density (average correlation between beliefs).

How many dimensions are there?

Using principal component analysis performed on each of the 70 belief systems separately, we find that the belief systems have 6-9 components with an eigen value >1. These components revolve around the themes/battery of questions included in the ESS survey (e.g., social benefits, climate change, migration, etc.). Following theoretical expectations, we are primarily interested in the 'meta-dimensions' that organize or unite these different themes. Specifically, the literature suggests that belief systems tend to be either one-dimensional (traditional left-right) or two-dimensional separating 'cultural' and 'economic' beliefs (Johnston and Ollerenshaw, 2020). Hence, in the next step we use factor analysis (minimal residual method with oblique oblimin rotation) focussing on one- and two-factor solutions. The mean explained variance across belief systems for the one-factor solutions is $M = 8.6$ per cent (SD = 2.8 per cent) and for the two-factor solutions it is $M_{\text{first factor}} = 7.9$ per cent (SD = 2.3 per cent) and $M_{\text{second factor}} = 6.1$ per cent (SD = 1.6 per cent). The mean number of beliefs that load stronger than 0.30 on these factors is $M = 5.7$ (SD = 2.7 per cent) for the one-factor solutions and $M_{\text{first factor}} = 5.0$ (SD = 1.7) and $M_{\text{second factor}} = 3.6$ (SD = 1.3) for the two-factor solutions. Hence, there is little support for one-dimensionality as a general organizing principle in these belief systems. The ranges also indicate that few, if any, belief systems come close to one-dimensionality as only five belief systems had a one-factor solution where more than 10

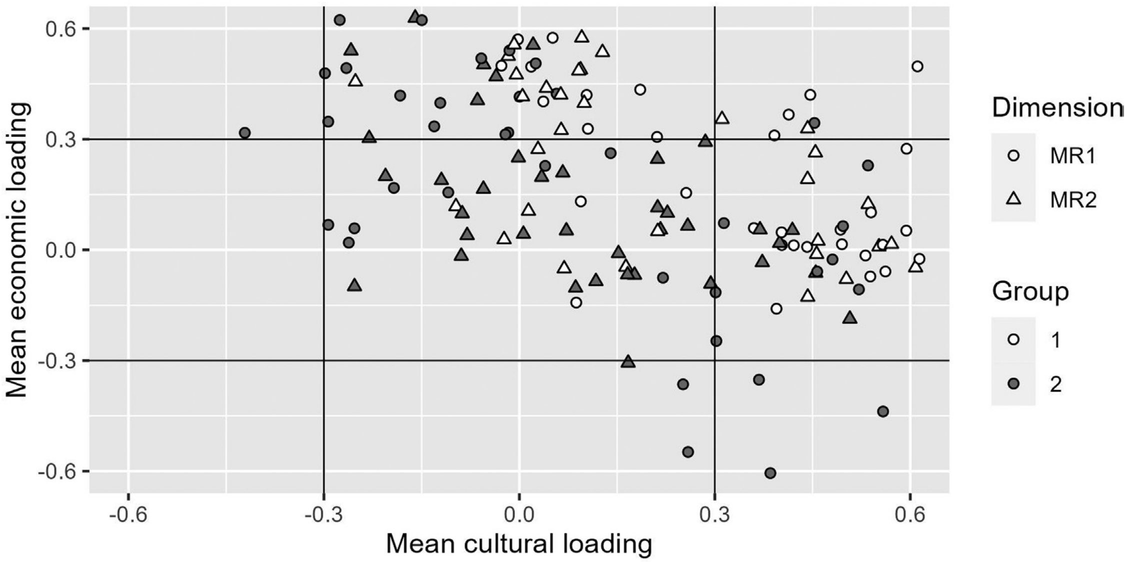

Figure 3 Scatterplot of mean cultural and economic loadings across the two dimensions of each belief system

beliefs have loadings >0.3. The maximum explained variance of these factors is 16.6 per cent. This suggests that the belief systems in Europe are almost all multi-dimensional, or even non-dimensional, as these factors explained little variance and united few beliefs. To further investigate to what extent these belief systems can be understood as containing an economic and cultural dimension, we explore the content of the factors from the two-factor solution.

What are these dimensions?

To gain a sense of which beliefs tend to go together in a dimension, we counted how often two beliefs have an absolute loading >0.3 on the same dimension. We find that the belief-pairs that most often have a high loading on the same factor are either two economic beliefs (e.g., anti-egalitarianism and anti-interventionism) or two cultural beliefs (e.g., anti-LGBT and anti-immigration). Belief-pairs consisting of an economic and a cultural belief are much less likely to both have a loading >0.3 on the same factor.[9] Though the exact ordering is different between the two groups, the pattern is the same. This suggests that these factors describe economic or cultural dimensions, rather than a dimension that contains a mix of economic and cultural beliefs.

To investigate this further, we focus on the loadings of two economic beliefs (*anti-egalitarianism* and *anti-interventionism*) and two cultural beliefs (*anti-LGBT* and *anti-immigration*). We have picked these beliefs as they have high loadings on the factors for both groups and are unequivocal economic or cultural beliefs. We classify a factor as *economic* or *cultural* if the mean loading of either of these pairs of beliefs is more than 0.3, and as *both* if the loadings of both pairs is more than 0.3. These mean loadings are displayed in the scatterplot in Figure 3 (summary in Table 3). 68 per cent of the factors were either economic, cultural, or both. A sizable amount, especially in Group 2 (41 per cent), cannot be classified either way. The factors classified as both contain some one-dimensionality, with respect to the economic and cultural beliefs. In seven of these factors, both pairs of beliefs load positively, such that right-wing cultural beliefs load with right-wing economic beliefs. In four factors, the pairs have opposing loadings: right-wing cultural beliefs load with left-wing economic beliefs. These last four are all in Group 2. 32 belief systems had one factor that was neither cultural nor economic. Four belief systems did not contain either a cultural or economic factor. Every country had at least one belief system with a factor classified as either cultural or economic.

Table 3 Summary of factors of all belief systems classified as none, economic, cultural, or both

	None	Economic	Cultural	Both
Group 1	16%	34%	41%	9%
Group 2	42%	29%	18%	7%
Total	30%	31%	29%	8%

Note: The table does not mention three factors that load negatively on the economic dimension and not on the cultural dimension. Percentages for the Group rows are as a proportion of all factors within each group (64 and 76, respectively). Percentages for the Total row are as a proportion of all (140) factors.

We supplement these analyses on the individual belief systems, with a factor analysis on the *average* correlation matrices of the belief systems of both broad groups. These are calculated by taking the average of every correlation in the correlation matrices for each individual belief system of each group. The factor analysis (performed similarly as for the individual belief systems) on the Group 1 average belief system shows an economic and a cultural factor (see Table 4). The factor analysis on the Group 2 average belief system shows a slightly different pattern. The first factor contains loadings above 0.3 for the two economic beliefs, but also a negative loading (–0.328) for anti-authoritarianism. The second factor contains a loading above 0.3 for anti-immigration, but not anti-LGBT with a loading of 0.240. Instead, it contains a loading above 0.3 for welfare chauvinism. As such, whereas in Group 1 we find the relatively 'traditional' economic and cultural dimensions, in Group 2 we find an 'authoritarian anti-egalitarian' dimension, and a nativism dimension revolving around anti-immigration and welfare chauvinism. The 'cultural' dimension in the average belief system for Group 2 revolves strongly around migration, rather than cultural values more broadly. However, this combination does not necessarily characterize more factors in Group 2 than the combination of anti-immigration and anti-LGBT. Analyzing the factor analyses of the individual belief systems, for either pair of cultural beliefs there are 18 factors that have an average loading above 0.3 (11 of these are the same factors). This might be an explanation for the larger diversity in Group 2: next to anti-immigration, both anti-LGBT and welfare chauvinism are important 'cultural' beliefs, but not necessarily all three simultaneously in each belief system.

Table 4 Factor analysis results for Group 1 and Group 2 average belief system

	Group 1		Group 2	
	MR1	MR2	MR1	MR2
	Cultural	Economic	Economic	Cultural
Anti-LGBT	**0.612**	–0.010	–0.066	0.240
Gender inequality	**0.528**	–0.047	0.020	0.064
Anti-immigration	**0.427**	0.080	–0.010	**0.576**
Anti-climate change renewables	0.296	0.124	0.186	0.095
Authoritarianism	0.241	–0.062	**–0.328**	–0.051
Climate scepticism	0.234	0.045	0.076	–0.015
Anti-libertarianism	0.216	–0.015	0.121	0.199
Euroscepticism	0.180	0.008	–0.063	0.210
Left-right identification	0.163	**0.357**	0.161	0.005
Anti-climate change taxes	0.161	0.001	–0.081	–0.007
Anti-climate ban of appliances	0.161	0.072	0.149	0.021
Benefits harm economy	0.160	**0.320**	0.169	–0.037
Welfare chauvinism	0.137	0.121	0.017	**0.396**
Benefits harm society	0.072	0.111	0.145	0.097
Anti-interventionism	0.009	**0.455**	**0.489**	0.031
Anti-social benefits (parents)	0.000	0.222	0.183	0.150
Educational spending	–0.042	0.094	0.033	–0.148
Anti-egalitarianism	–0.053	**0.592**	**0.446**	–0.105
Anti-basic income	–0.070	0.212	0.250	0.051
Anti-social benefits (low income)	–0.180	0.069	0.036	0.039
SS loadings	1.286	0.956	0.811	0.718
Proportion Var	0.064	0.048	0.041	0.036
Cumulative Var	0.064	0.112	0.041	0.076
Factor correlations (MR1–MR2)	0.241		–0.202	

Note: Factor loadings > 0.3 are in bold. Factors were rotated (oblique oblimin). Cultural/economic are factor names. Beliefs are sorted by column 2 (Group 1 cultural).

How do these dimensions relate to each other?

To explore how the dimensions from all analyses relate to each other, we calculate three correlations: (i) we constructed two scales (arithmetic mean) with the above-mentioned two economic beliefs (anti-egalitarianism and anti-interventionism) and two cultural beliefs (anti-LGBT and anti-immigration), and we calculate for each group of respondents that share a belief system the correlation between the two scales ($r_{\text{Group 1}} = 0.09$, $r_{\text{Group 2}} = -0.11$). (ii) We take the factor correlations between the first two factors extracted in the factor analyses on the individual belief systems (including the factors not classified as either cultural or economic; $r_{\text{Group 1}} = 0.14$, $r_{\text{Group 2}} = -0.03$). And (iii) we take the factor correlations between the first two factors extracted in the factor analyses on the average belief systems ($r_{\text{Group 1}} = 0.24$, $r_{\text{Group 2}} = -0.20$). In the case of the first and second correlations, these are the mean correlations across the belief systems.

These correlations are all positive for Group 1 and negative for Group 2. Thus, in Group 1 right-wing cultural beliefs tend to go together with right-wing economic beliefs, similar to a (traditional) one-dimensional logic. In Group 2 right-wing cultural beliefs tend to go together with left-wing economic beliefs, similar to an alternative cultural-economic two-dimensional logic. However, these correlations are small. In particular, the mean factor correlation on the individual belief systems for Group 2 is small, only −0.03. This is in line with the larger number of factors not being classified as economic or cultural dimensions. If these factors revolve around a different logic than economic or cultural, the relation between the two factors is likely different and, as the low correlation shows, not consistently in a specific direction across the individual belief systems.

Summary

We find that European countries contain multiple belief systems that differ both within and across countries. These belief systems can be grouped into two broader groups of belief systems. These groups did not differ in mean constraint as might be expected. Instead, the differences between these groups lie, firstly, in how homogenous these groups were. The belief systems in Group 1 were more similar to each other than the belief systems in Group 2 were similar to each other. Secondly, there was little evidence of one-dimensionality, and the factors (dimensions) that were extracted from the factor analyses explained little variance and united few beliefs for most belief systems. A large amount of the main two dimensions across the subnational belief systems can be characterized as either economic or cultural, though this does not hold for all belief systems—especially in the more diverse Group 2. Lastly, the economic and cultural dimensions tended to correlate positively in Group 1 and negatively in Group 2, though these correlations were not high.

Who shares these belief systems?

We explored how the people who share Group 1 and 2 belief systems differ demographically. To that end, we ran multilevel logistic regression analyses with Group 1 (vs. Group 2) membership as the dependent variable and demographic indicators as independent variables (see Supplementary Table A4 for details on variable measurement). We excluded eleven countries where all belief systems are in the same group, and thus have zero variance on the dependent variable. Analyses proceed with the twelve countries that have belief systems in both groups ($N = 17{,}939$). Results show that all demographic variables are significantly related to Group 1 vs. Group 2 membership (full results listed in Supplementary Table A6). We plotted predicted values in Figure 4. Individuals in the belief systems in Group 1 are more likely to be higher educated, have a higher income, be female, be younger, more secular, less likely to be part of an ethnic minority group, and more likely to live in a big city or its suburbs than individuals that share belief systems in Group 2. Comparing the size of the differences per variable shows that people's educational attainment is most strongly related to Group 1 membership. Since these analyses exclude eleven countries, we also analyzed the relationship between demographics and belief system membership for each belief system separately (see Supplementary Table A7). The results of these analyses show that education is the primary predictor of belief system membership. For instance, higher education (compared to lower education) was significantly related to membership for 67 per cent of belief systems and only in two countries higher education was not significantly related to membership of any belief system. Other demographics had significant relations for around 40 per cent of belief systems, with urbanization and ethnic minority as exceptions with 59 per cent and 21 per cent significant relations.

Another main demographic difference between the two groups is that Group 1 has more belief systems from Western European countries, and Group 2 has more belief systems from Eastern European countries or Israel. In fact, Group 1 only contains five belief systems from Eastern European countries or Israel (16 per cent). Conversely, only thirteen belief systems in Group 2 are from Western European countries (34 per cent).

Are belief systems related to voting behaviour?

We find that there are multiple European belief systems and that they differ in demographic membership.

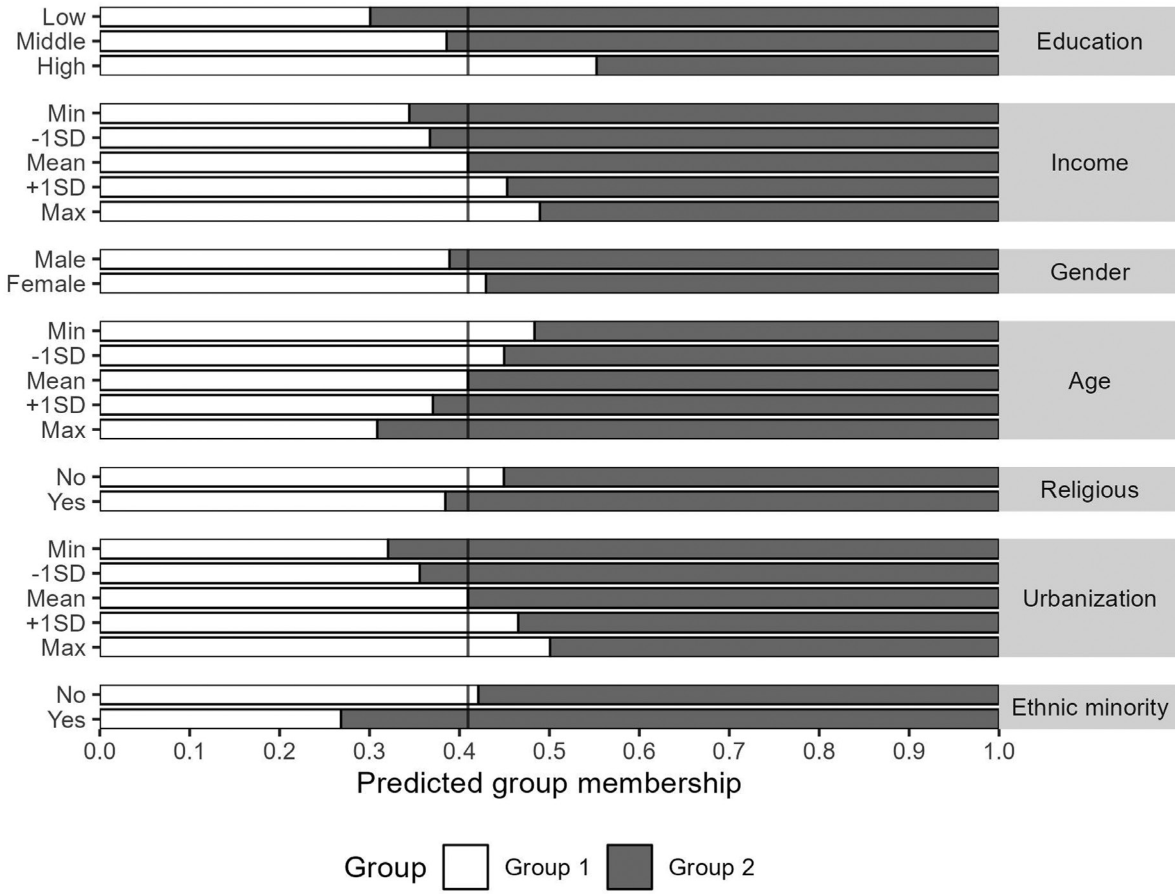

Figure 4 Predicted values of group membership across demographic variables in countries with belief systems in both groups

Note: Income, age, and urbanization are standardized, other variables are categorical. Predicted values based on a multilevel linear regression analysis, see Supplementary Table A6. The vertical line indicates the mean Group membership (0.41 for Group 1, 0.59 for Group 2). The countries with belief systems in both groups are as follows: Austria, Belgium, Estonia, France, Germany, Ireland, Israel, Italy, Poland, Slovenia, Switzerland, and United Kingdom.

However, are these different belief systems related to behavioural outcomes such as voting preferences? We estimated a multilevel multinomial logistic regression with self-reported voting behaviour in the last national election as the dependent variable. In Figure 5, we have illustrated the predicted probabilities for each contrast with all demographics variables set to their mean value (full results available in Supplementary Tables A8–A10). In order to account for the variation in political parties in different countries, we have grouped the voting behaviour variable in three general types of voting: loyalty voting, voice voting, and exit voting (Hirschman, 1970). Loyalty voting means voting for a mainstream party,[10] voice voting means voting for a non-mainstream party (which is subdivided in five categories: far right, far left, populist, populist far right, or populist far left, according to the Popu-List classification; Rooduijn *et al.*, 2019), and exit voting means vote abstention.

The results show that being a member of a belief system in Group 1 (rather than Group 2) is associated with more loyalty voting, and less voice and exit voting. Importantly, the association between-belief system structure and voting behaviour exists over and above the demographic factors included in the model. This suggests that belief system structure provides additional information above and beyond demographic factors. Although, for voice voting, we only see a difference between-belief system groups for populist far-right voting, specifically. Other categories of voice voting are not significantly different between the two groups. This is in line with earlier research that showed that belief system group membership is related to populist far-right voting vs. mainstream voting (Daenekindt, De Koster and Van der Waal, 2017). We demonstrate also that this does *not* extend to other forms of 'voice' or protest voting.

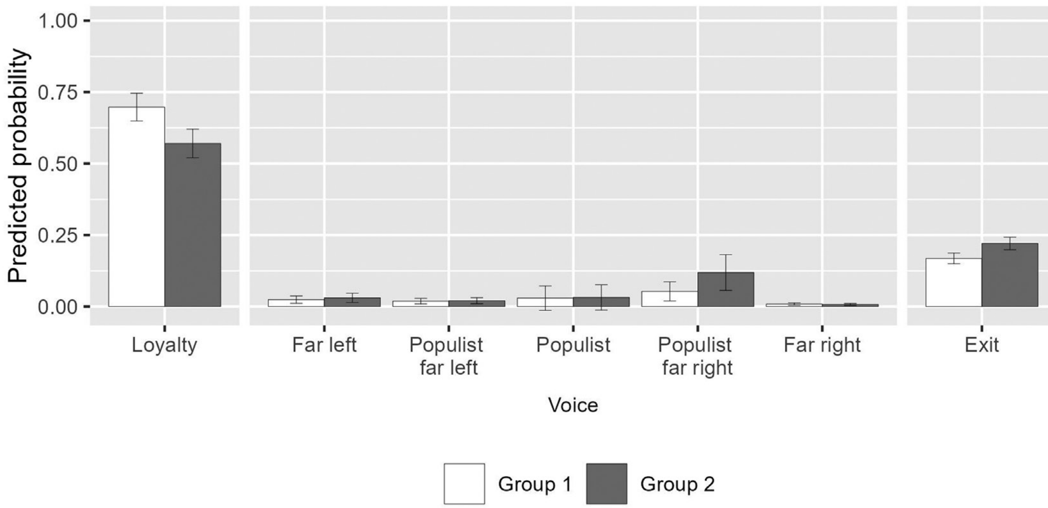

Figure 5 Predicted probabilities of vote behaviour across Groups 1 and 2

Note: Error bars denote 95 per cent confidence intervals. Differences between groups (within a vote behaviour category) are significant for Loyalty, Populist far right, and Exit ($P < 0.001$). Full results, including exact predicted means and testing of differences, are listed in Supplementary Tables A7–A9.

General discussion and conclusion

Research on belief systems has largely focussed on single countries and comparisons across countries have been limited. We investigated the structure of the belief systems of 23 European countries. We used correlational class analysis (CCA) to uncover the most prevalent belief systems in each country. In order to investigate what belief systems in European societies, and those who hold them, have in common, we asked three questions: (i) are the structures of the belief systems similar across countries in Europe? (ii) Which social groups are likely to have similar belief systems? (iii) How are different belief systems related to voting behaviour?

Our findings revealed two groups of belief systems in Europe which differ in at least four crucial ways: (i) Group 1 portrayed more homogeneity in belief systems across countries. (ii) Belief systems, especially Group 2, often did not contain (either) a cultural or an economic dimension, and to the extent they did, these dimensions explained little variance and united few beliefs. They tended to correlate positively in Group 1 and negatively in Group 2. (iii) Group 1 belief systems were shared, among others, more among the higher educated, the rich, and in Western Europe. (iv) Group 2 membership was related to less mainstream and more populist far-right voting. In the following paragraphs, we discuss these findings and their implications in further detail.

First, Group 1 portrayed more homogeneity in belief systems across countries. This is an aspect of belief systems that previous research (that focussed on single countries) could not find and hints at how elite political discourse is organized internationally. For instance through the international spread of the modern educational system and universities, which provides a socialization context for especially higher educated people (Schofer and Meyer, 2005). Those who are more likely to share an alternative belief system (e.g., Group 2) are perhaps less likely to populate international institutions that can provide such international coordination, leading to less similarity in belief systems across countries. This might also reflect the relative novelty of the belief system logic found in Group 2, as the traditional left-right belief system is historically a more dominant way of thinking about politics (Caprara and Vecchione, 2018). Homogeneity in 'Group 2' might increase over time with increasing international coordination of, for instance, radical right politics on social media (Froio and Ganesh, 2019), or in international political organizations and groups, such as Identity and Democracy in the European Parliament (see also McDonnell and Werner, 2020) and 'National Conservatism' of the Edmund Burke Foundation.

Second, we identified per belief system which two dimensions were most pronounced and whether these can be understood as cultural and economic dimensions. We found that many, but not all, of these dimensions can be understood as cultural or economic, especially in Group 1. In Group 2, results were more mixed as dimensions were less clearly cultural or economic. This is likely due to a larger diversity in Group 2 that makes it more difficult to make clear conclusions of the meaning of the belief systems in this group (see also point four, below). It seems that in Group 2 the 'cultural' dimensions are less likely to unite a broad range of cultural beliefs, but rather have either migration/welfare chauvinism or LGBT issues as a central belief. The first could perhaps be understood as revolving around nativism, the latter as (gender) traditionalism. In Group 1, these cultural beliefs were more likely to load on a single dimension. We also found that in Group 1 the economic and cultural dimensions were more likely to correlate positively, that is, if a person was right-wing on one belief dimension, that person was likely to also be right-wing on the other belief dimension. As such, the belief dimensions go together in a way that is similar to the traditional left-right political continuum. Group 2 dimensions were more likely to correlate negatively, similar to the 'alternative' belief system described by earlier literature (Baldassarri and Goldberg, 2014). However, in both groups, these two dimensions united only a few of the 20 beliefs, i.e.: little variance, and, importantly, did not correlate strongly. As such, it is difficult to conclude that all Group 1 belief systems are ideologue belief systems and all Group 2 belief systems alternative. Though the groups are likely to contain, respectively, ideologue and alternative belief systems.

Third, differences between the two groups of belief systems were also in terms of *who* shares which belief system. We found that those who shared a belief system in Group 1 were more likely to be higher educated, have a higher income, be female, be younger, more secular, less likely to be part of an ethnic minority group, more likely to live in a big city or its suburbs, and be from Western Europe than individuals that shared a belief system in Group 2. Education was an especially important factor in predicting belief system group membership (independently from other demographic factors). Only in two countries, higher education was not related to membership of any belief system. This underscores the importance of how education is a structuring factor in modern politics (Easterbrook, Kuppens and Manstead, 2016) and is consistent with findings that educational level has been linked to political events such as Brexit and the election of Trump in 2016 (Hobolt, 2016; Silver, 2016). Education could function as a proxy for cultural capital (Van der Waal, Achterberg and Houtman, 2007), or as an important identity and basis for societal status (Spruyt and Kuppens, 2015; Van Noord *et al.*, 2019, 2023; Sandel, 2020).

Education is also often seen as a proxy for political skills or engagement (Verba, Schlozman and Brady, 1995). However, this is at odds with our results indicating that Group 2 belief systems contained on average highly similar mean belief system density as belief systems in Group 1 (despite being shared by relatively less educated individuals). This is not to say that there were no differences in density among the belief systems, rather these differences were found *within* the two groups, instead of *between* the two groups. As such, while Group 1 showed a more typical ideologue correlational structure, the distinction between Groups 1 and 2 is perhaps only partly a difference in political engagement. This aligns with the work of Groenendyk, Kimbrough and Pickup (2020), who show that people can meaningfully and strategically deviate from the politically elites and thus that non-typical belief systems do not have to signal a lack of political sophistication. Further research should investigate to what extent the higher density among these belief systems in Group 2 are indeed indications of political sophistication.

Fourth, we found that Group 1 membership was associated with more *loyalty* voting (voting for liberal-democratic/establishment parties) and Group 2 membership with more *voice* voting (though only in the form of populist far-right voting) and *exit* voting (vote abstention). This indicates that this alternative belief system is not so much an anti-establishment belief system per se, and thus appealing to all forms of voice voting, but in contemporary Europe rather more closely connected to specifically populist far-right politics. We do not interpret the relationship between-belief system membership and voting behaviour to be causal, though it is possible that aspects of belief systems are causally related to behaviour. Though, primarily we have aimed to show in a general sense how the belief systems identified here are related to voting behaviour. Further research can elaborate on these associations and (aspects of) belief systems. These associations with voting behaviour in the context of Group 2 belief systems being more likely to be from Eastern European countries also reflect the generally stronger success of radical right-wing populist parties (whose politics are also more in line with a two-dimensional cultural-economic structure, see, e.g., Daenekindt, De Koster and Van der Waal, 2017) in Eastern European countries such as Poland and Hungary, compared to Western European countries. Furthermore, while the alternative belief system has been observed before (e.g., Lipset, 1959)—only with the recent rise of populist far-right politics has this found political articulation. Indeed, while belief systems may both follow and shape elite discourse, the finding that Group 2 membership is related to populist far-right voting may be seen as evidence for a successful elite political articulation of politics alternative to the traditional one-dimensional left-right political axis.

That belief systems in Group 2 were more likely to contain a two-dimensional system, but also more likely to be from Eastern Europe, implies that *European* politics are not solely about different beliefs, but also about different belief systems. We did not find very explicitly the existence of 'unstructured' belief systems, as these belief systems did not make up a separate group. What we did find were differences in density/structuredness within the two groups, indicating that the belief systems that are less structured are present in both groups. To the extent, we find less- or unstructured belief systems, they still seem to be structured around traditional left-right or alternative logics. As such, while previous research found separate types of belief systems, we find that European belief systems seem to differ *gradually* from each other, with two structured 'ideal types' of belief systems at either pole and the relatively less- or unstructured belief systems in the middle, often tending towards one-dimensional or alternative structures, with only few, if any, truly unstructured belief systems.

While our robustness tests show that our results are generally robust for both selection of items and selection of respondents (see Supplemental Material), the current study also has limitations. Firstly, the selection of countries and beliefs was determined by the availability in the dataset. This means that we only analysed a relatively low number of Eastern European countries (6 of 23). And while belief systems should have validity beyond the (broad) selection of beliefs, and current results were largely replicated in a different ESS wave (NSD, 2008), results remain (partly) dependent on item wording and selection of themes of beliefs. Secondly, CCA does not take asymmetrical correlations into account, where two beliefs are only related for those who score high on one of these beliefs, but these beliefs are not or only weakly related for those who score low on this belief. The result of this would be that people with opposing beliefs are more likely to be seen as having different belief systems, when they, in reality, share one that contains asymmetrical belief-relations. It is difficult to gauge the extent to which this is the case, though other methods to estimate belief systems using panel studies focussing on dynamic constraint (Turner-Zwinkels and Brandt, 2022) or individual-level estimates of belief systems (Brandt, 2022) could circumvent this problem. Furthermore, while CCA groups individuals together based on similarity, some individuals might possess belief systems that are idiosyncratic and thus different from more widely shared belief systems that are identified by CCA (while still being potentially grouped together). As we cannot estimate the belief systems of specific individuals in the sample (but see Brandt, 2022), it is also difficult to gauge how many people possess such belief systems, and how it affects the estimation of group-level belief systems. Lastly, while our bottom-up approach allows us to find empirical patterns that top-down analyses (e.g., hypothesis-testing) might not find, theory building would benefit from further research that uses methods more focussed on hypothesis-testing and that could demonstrate the robustness of our findings.

In sum, we used bottom-up methods to analyse political belief systems that exist in different countries in Europe. We demonstrate that Europe contains a wide variety of belief systems that differ in important ways from each other. Within this wide variety, we found two main groups of belief systems that differ primarily in how cultural and economic dimensions relate to each other (either positively or negatively). Though differences within these groups were found in the strength of correlations between beliefs, there was no difference between these groups. While almost all belief systems displayed either an ideologue or alternative structure, the structured belief systems were less structured than expected. At the same time, even the least structured belief systems tended to display patterns similar to the ideologue or alternative belief systems, and were thus not as unstructured as expected. Thus, few belief systems were either very structured or very unstructured. Membership of belief systems where cultural and economic dimensions correlated negatively was (primarily) negatively associated with educational level, and with more populist far-right voting and vote abstention.

Notes

1. Data and scripts are available at Open Science Framework: https://osf.io/hc3yz/.
2. We tested for the robustness of our results in the 2008 wave of the ESS in Supplemental Material.
3. CCA also only provides a single unambiguous solution, rather than multiple solutions out of which researchers have to choose the most appropriate as is common in FA or LCA.
4. One might expect that we analyse the whole dataset with a single CCA, but this *assumes* that there are similar European belief systems and finding country-specific belief systems using this approach is unlikely. Such an approach would then not provide much information on what lies 'beneath' these European belief systems.
5. We have labelled the belief systems either 'country #' or 'country-code #'. The numbers of belief systems (#) have no substantive meaning and simply reflect the order in which they were found by CCA. Country codes are based on those used in ESS.
6. This excludes so called degenerate classes, that consist of single individuals that had no significant correlations with any other individual.
7. Following earlier research (Boutyline, 2017; Boutyline and Vaisey, 2017; Daenekindt, 2019), we use SEM to test, per country, whether the different belief systems give a better fit than a single population-level correlation matrix. In all countries, separate belief systems give a better fit with Chi² test and AIC, but with BIC in 7 of 23 countries the single matrix is preferred over the different belief systems. Each of these seven countries contain either 3 or 4 belief systems. See Supplementary Appendix 4 for more details.
8. The number of groups is determined by the algorithm. We explore other clustering algorithms in Supplementary

Appendix 3. These found no substantive differences from the results presented here.

9. The most occurring economic-cultural belief pair with both loadings >0.3 is anti-immigration–anti-egalitarianism for Group 2, which occurs 10 times. Anti-egalitarianism–anti-interventionism occurs 20 times in Group 2, and there are 13 belief-pairs that load >0.3 on the same factor more often than anti-immigration–anti-egalitarianism. For Group 1, this is anti-immigration–benefits hurts economy with 10 high loadings on the same factor, though there are 19 belief-pairs that do not combine a cultural with an economic belief that occur more often than that.
10. Since voice vote is based on far right/far left/populism, this might mean that in those countries where these parties are in power, voice might entail voting for a party that is currently in government.

Supplementary data

Supplementary data are available at *ESR* online.

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Data availability

Data and replication files are available at https://osf.io/hc3yz/.

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Jochem van Noord is a post-doctoral researcher at the Vrije Universiteit Brussel. His interests revolve around how social inequality, mainly around class and education, affect political and social behaviour and attitudes. For this, he gives special attention to processes of group conflict, identity, and social status. He has recently published in *Social Forces*, *Personality and Social Psychology Bulletin*, and *Political Geography*.

Felicity Mary Turner-Zwinkels, Post-doctoral research, Department of Sociology, Tilburg University. My research explores political identity and political belief systems and how they influence people. I do so by exploring questions such as; how does the structure of political beliefs influence how attitudes change or how we relate to fellow ingroup or outgroup members? Through my work, I aim to deepen our understanding of both the content and structure of political identity and political belief systems and how and why they impact political behaviour.

Rebekka Kesberg is a post-doctoral researcher at the University of Sussex and University of Amsterdam. Her research focuses on democratic attitudes and how they relate to political violence and democratic backlash.

Mark J. Brandt is an assistant professor in the Department of Psychology at Michigan State University. He is interested in how belief systems work.

Matthew J. Easterbrook is a Reader in Social Psychology at the University of Sussex, UK. His research investigates educational, economic, and political inequalities, and how we can intervene to reduce them.

Toon Kuppens is an assistant professor at the Faculty of Behavioural and Social Sciences of the University of Groningen. His research focuses on psychological aspects and consequences of socioeconomic status, specifically education. He investigates the intergroup relation between education-based groups, for example attitudes towards lower and higher educated people, and the stigma of being less educated.

Bram Spruyt is associate professor of Sociology at the Vrije Universiteit Brussel. His main research interests include public opinion research, the sociology of education and social identities. In the past years, he has published extensively on topics such as populism, education-based status, and feelings of misrecognition.